

P2559R0

Plan for Concurrency Technical Specification Version 2

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Source:

[GitHub](#)

Issue Tracking:

[GitHub](#)

Project:

ISO/IEC JTC1/SC22/WG21 14882: Programming Language — C++

Audience:

WG21

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§ 1. Introduction

We've been slowly filling the Concurrency Technical Specification (TS) version 2 for approximately six years. It's time to move this work to publication so we can get field experience with the contents of

the TS.

§ 2. Scope

The following content is already in the Concurrency TS v2 Working Draft [\[N4895\]](#):

- [\[P1121R3\]](#): Hazard Pointers
- [\[P1122R4\]](#): Read-Copy-Update (RCU) Data Structures

The following content has been approved by the Concurrency and Library Evolution groups for the Concurrency TS v2 and is awaiting Library review:

- [\[P0290R2\]](#): `synchronized_value`
- [\[P0561R6\]](#): `<experimental/snapshot>` - Deferred Reclamation RAII Interface
- [\[P1478R7\]](#): Byte-wise Atomic `memcpy`
- [\[P2396R0\]](#): Concurrency TS 2 Fixes

The following content has been approved by the Concurrency group for the Concurrency TS v2 and is awaiting Library Evolution review:

- [\[P1202R4\]](#): Asymmetric Fences

The following content has not yet been reviewed by the Concurrency group, but it should be included in the Concurrency TS v2:

- [\[P2535R0\]](#): Message Fences

The following content has been approved by the Concurrency group for the Concurrency TS v2 and was initially reviewed by Library Evolution years ago, but revisions are needed and the content is no longer being actively developed. This content should not be included in the Concurrency TS v2:

- [\[P1958R0\]](#): Concurrent Buffer Queue
- [\[P0260R4\]](#): Concurrent Queues
- [\[P0261R4\]](#): Distributed Counters

§ 3. Timeline

- 2022-2H: Complete Concurrency and Library Evolution review of all Concurrency TS v2 content.
- 2023-1H: Complete Library review of all Concurrency TS v2 content.
- 2023-2H: Publish the Concurrency TS v2.

§ References

§ Informative References

[N4895]

Michael Wong. *Working Draft, Extensions to C++ for Concurrency Version 2*. 15 August 2021. URL: <https://wg21.link/n4895>

[P0260R4]

Lawrence Crowl, Chris Mysis. *C++ Concurrent Queues*. 13 January 2020. URL: <https://wg21.link/p0260r4>

[P0261R4]

Lawrence Crowl. *C++ Distributed Counters*. 13 January 2020. URL: <https://wg21.link/p0261r4>

[P0290R2]

Anthony Williams. *apply() for synchronized_value<T>*. 2 March 2017. URL: <https://wg21.link/p0290r2>

[P0561R6]

Geoffrey Romer, Andrew Hunter. *An RAII Interface for Deferred Reclamation*. 2 February 2022. URL: <https://wg21.link/p0561r6>

[P1121R3]

Maged M. Michael, Michael Wong, Paul McKenney, Geoffrey Romer, Andrew Hunter, Arthur O'Dwyer, D. S. Hollman, JF Bastien, Hans Boehm, David Goldblatt, Frank Birbacher, Mathias Stearn, Jens Maurer. *Hazard Pointers: Proposed Interface and Wording for Concurrency TS 2*. 9 April 2021. URL: <https://wg21.link/p1121r3>

[P1122R4]

Paul E. McKenney, Paul McKenney, Michael Wong, Maged M. Michael, Geoffrey Romer, Andrew Hunter, Arthur O'Dwyer, Daisy Hollman, JF Bastien, Hans Boehm, David Goldblatt, Frank Birbacher, Erik Rigtorp, Tomasz Kamiński, and Jens Maurer. *Proposed Wording for Concurrent Data Structures: Read-Copy-Update (RCU)*. 14 May 2021. URL: <https://wg21.link/p1122r4>

[P1202R4]

David Goldblatt. *Asymmetric Fences*. 15 February 2022. URL: <https://wg21.link/p1202r4>

[P1478R7]

Hans Boehm. *Byte-wise atomic memcpy*. 5 February 2022. URL: <https://wg21.link/p1478r7>

[P1958R0]

Lawrence Crowl. *C++ Concurrent Buffer Queue*. 13 January 2020. URL:
<https://wg21.link/p1958r0>

[P2396R0]

David Goldblatt. *Concurrency TS 2 fixes*. 14 June 2021. URL: <https://wg21.link/p2396r0>

[P2535R0]

Olivier Giroux, Daniel Lustig. *Message fences*. 2 February 2022. URL: <https://wg21.link/p2535r0>